

Integrating Generative AI in Cybersecurity Education: Case Study Insights on Pedagogical Strategies, Critical Thinking, and Responsible AI Use

Project Overview

The document explores the integration of Generative Artificial Intelligence (GenAI) in education, particularly within the realm of cybersecurity, showcasing its potential to significantly enhance students' critical thinking, regulatory awareness, and problem-solving abilities. It presents a structured framework for effectively embedding GenAI tools in educational settings, utilizing a two-stage implementation approach that includes tutorial exercises and assessments. Findings from the study reveal that the incorporation of GenAI leads to improved student engagement and the development of analytical skills. However, it also identifies challenges such as the risk of over-reliance on AI technologies and varying levels of AI literacy among students. To address these issues, the document advocates for a balanced educational approach that combines the advantages of AI assistance with the necessity of human expertise, ensuring that students benefit from both innovative technologies and traditional learning methods. Overall, the document underscores the transformative potential of GenAI in education while calling for careful consideration of its limitations.

Key Applications

Use of GenAI tools like ChatGPT for generating cybersecurity policies and assessments.

Context: Cybersecurity education for undergraduate and postgraduate students.

Implementation: A two-stage approach embedding GenAI in tutorials for critique and refinement of AI-generated content, followed by assessments requiring application of AI outputs to real-world scenarios.

Outcomes: Enhanced critical thinking, improved regulatory awareness, and better alignment of security policies with industry standards.

Challenges: Over-reliance on AI outputs, variability in student proficiency with AI tools, and the need for contextual customization of AI-generated content.

Implementation Barriers

Pedagogical Challenge

Students may develop over-reliance on AI-generated content, treating it as authoritative without adequate critique. Additionally, differences in student familiarity with AI tools can lead to uneven engagement and learning outcomes.

Proposed Solutions: Incorporating structured reflection exercises and explicit prompts to emphasize the limitations of AI outputs. Providing additional scaffolding and targeted tutorials for students less experienced with AI.

Context-Specific Customization

AI-generated content often lacks specificity needed to align with organizational contexts and regulatory frameworks.

Proposed Solutions: Encouraging research and iterative refinement to ensure alignment with industry standards.

Logistical Constraints

Challenges in coordinating real-world business engagements for student assessments.

Proposed Solutions: Offering structured templates for interviews and alternative case study options.

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